EPM Planning and Forecasting Data in EGDL (Programmatic Access)

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Overview

The data export process involves transferring financial and planning data from **Oracle EPM Cloud Planning** to **Expedia Group Data Lake (EGDL)**. The process is automated using scheduled jobs, Oracle Cound Integrations, and cloud storage.

Oracle Integration Cloud (**OIC**) enables automated data export from **Oracle EPM Cloud** to **AWS S3 EG Data Lake** by orchestrating data extraction, transformation, and storage processes. This integration helps maintain backups, enable advanced analytics, and ensure seamless data movement between cloud platforms.

Current State:

- Today, we provide Actual and Forecast data extracts to various teams (TPSP, Marketing Analytics Godzilla, eCP, LPS)through multiple methods.
- Some teams receive data via **direct push to SQL Server**, while others rely on **file-based extracts**.
- Additionally, some data is already being **published to the Expedia Group Data Lake (EGDL)**.
- This fragmented approach requires **multiple integration points** and **manual intervention**, leading to inefficiencies and potential inconsistencies.

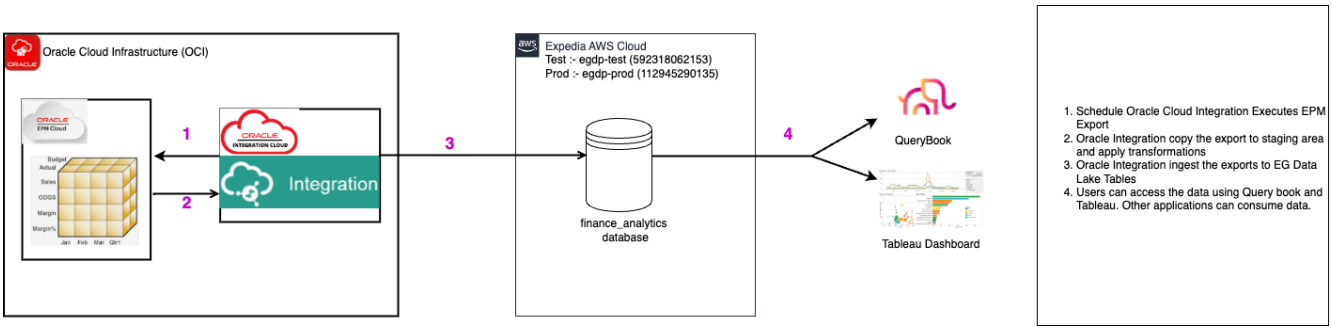
- On Prem EPM Outbound - Call Center/GCO data
- On Prem EPM Outbound - Channel Extract to EGDL
- On Prem EPM Outbound - eCP Data Extract
- On Prem EPM Outbound - LPS Data Extract

Future State – Unified Approach:

- In the new system, we will **standardize** how EPM data is shared by publishing **both forecast and actual data** for all teams to EGDL.
- This shift eliminates the need for **custom data extracts** and **manual transfers**, providing a **single source of truth** for financial data.

High-Level Architecture

1. **Trigger**: The OIC integrations are triggered via a scheduled job in OIC.
2. **Extract Data from Oracle EPM**: OIC invokes the **EPM Cloud Adapter** to execute a **Data Management export jobs** that generates a file (CSV, ZIP, or DAT format).
3. **Download Exported File**: The generated file is retrieved from EPM's **Outbox** folder using the **EPM Adapter** or REST API.
4. **Transformations:** By application and use case required **transformations** will be performed while transforming data from **EPM outbox** to **EG Data Lake**.
5. **Upload to AWS S3**: The file is sent to **AWS S3 EG Data Lake** using the **Amazon S3 Adapter** or REST API.
6. **Data Access**: Consumer applications can access the data from EGDL by registering the access pattern via EG **Security groups**
7. **Monitor & Notify**: The integration logs success or failure and notifies relevant stakeholders. The status of schedules and jobs are tracked in the OIC dashboards.



Data Access and Security

The data publishing process can support **dimension-based security**, allowing access restrictions based on user roles and data attributes, such as brand, product or account.

Hive Tables Access Link

- **Ranger Policy:**
  - A **Ranger policy** defines **who** can access specific resources and **what** actions they can perform within **Apache Ranger**.
- **Table-Level Access:**
  - Restricts access at the **table level**, ensuring only authorized users can view specific datasets.
- **Column-Level Access:**
  - **Column restrictions** and **column masking** control the visibility of sensitive information based on user roles.
- **Row-Level Access:**
  - **Row-level filtering** enforces access restrictions at a granular level, limiting data visibility based on data attributes such as brand, product, or account.

EPM data security in EGDL:

We have implemented a security model(**Row-Level Access**) to grant appropriate EPM cloud data access to the consumers in EGDL test and prod environments. For this we've created 3 security groups -

- **epm-access-all** → Access to both b2b and b2c brands
- **epm-access-b2c** → Access to only b2c brands
- **epm-access-b2b** → Access to only b2b brands
- Gishee's team, Adriana's team and Gogul have been put into **epm-access-all**
- Godzila team has been put into **epm-access-b2c**
- We will expand these groups as we move on.
- 'epm-access-b2c' and 'epm-access-b2b' are semi private groups, so one should be able to request access to these and we will review and approve it. If you think you need access to epm-access-all group (private), please reach out to @amarathe or `@Hari Gopal Anil` so that we can add you. Note that you should be part of only one SG.
- There is no security in place for fxrates table.

Row level filters -

- fin_plan (prod) - https://github.com/eg-internal/egdl-access-control/blob/master/egdp-prod/policies/rowFilters/epm_fin_plan_userpool_access.json
- fin_plan (test) - https://github.com/eg-internal/egdl-access-control/blob/master/egdp-test/policies/rowFilters/epm_fin_plan_userpool_access.json
- vtr_plan (prod) - https://github.com/eg-internal/egdl-access-control/blob/master/egdp-prod/policies/rowFilters/epm_vtr_plan_userpool_access.json
- vtr_plan (test) - https://github.com/eg-internal/egdl-access-control/blob/master/egdp-test/policies/rowFilters/epm_vtr_plan_userpool_access.json

Airflow jobs:

Access: Join `sg-fde-plan-forecasting` security group to get access to the below airflow jobs - askEG Login Page - askEG

- **FIN_PLAN**:
  - **Prod**:
    - Active forecast - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.dw.exp-aws.net/dags/epm_finplan_active_forecast_export_task/grid
    - Actuals - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.dw.exp-aws.net/dags/epm_finplan_actuals_export_task/grid
    - Archived forecast (adhoc) - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.dw.exp-aws.net/dags/epm_finplan_archived_forecast_export_task/grid
    - Plan - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.dw.exp-aws.net/dags/epm_finplan_cube_data_export_task/grid
  - **Test**:
    - Active forecast - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.test.exp-aws.net/dags/epm_finplan_active_forecast_export_task/grid
    - Actuals - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.test.exp-aws.net/dags/epm_finplan_actuals_export_task/grid
    - Archived forecast (adhoc) - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.test.exp-aws.net/dags/epm_finplan_archived_forecast_export_task/grid
    - Plan (adhoc) - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.test.exp-aws.net/dags/epm_finplan_cube_data_export_task/grid
  - **Airflow variables for both Prod and Test environments:**
    - **Active forecast:**

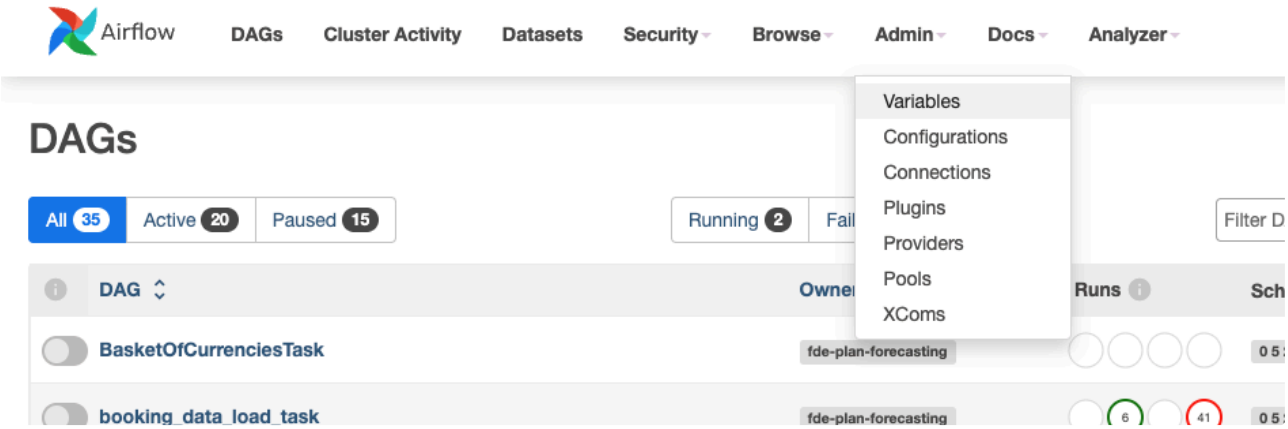
        - “fin-plan-data-load-image-version” → docker image version (github commit sha)
        - “fin-plan-active-forecast-scenario” → Forecast
        - “fin-plan-active-forecast-version” → Working_Prior
        - “fin-plan-active-forecast-years” → FY26,FY27
        - “fin-plan-active-forecast-period” → Yeartotal
    - **Actuals:**
        - “fin-plan-data-load-image-version” → docker image version (github commit sha)
        - “fin-plan-data-load-actual-scenario” → Actual
        - “fin-plan-data-load-actual-version” → Final
        - “fin-plan-data-load-actual-years” → FY26
        - “fin-plan-data-load-actual-period” → Feb
    - **Archived forecast (adhoc):**
        - “fin-plan-data-load-image-version” → docker image version (github commit sha)
        - “fin-plan-archived-forecast-scenario” → Feb Forecast
        - “fin-plan-archived-forecast-version” → FY26_Archive
        - “fin-plan-archived-forecast-years” → FY26,FY27
        - “fin-plan-archived-forecast-period” → Yeartotal
    - **Plan (adhoc):**
        - “fin-plan-data-load-image-version” → docker image version (github commit sha)
        - “fin-plan-data-load-scenario” → Plan
        - “fin-plan-data-load-version” → Final
        - “fin-plan-data-load-years” → FY26
        - “fin-plan-data-load-period” → Yeartotal
  - Note: Use period as “Yeartotal” if you want to run it for all months.
- **VTR_PLAN**:
  - **Prod**:
    - Active forecast - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.dw.exp-aws.net/dags/epm_vtrplan_active_forecast_export_task/grid
    - Plan & Archived forecast (adhoc) - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.dw.exp-aws.net/dags/epm_vtrplan_cube_data_export_task/grid
  - **Test**:
    - Active forecast - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.test.exp-aws.net/dags/epm_vtrplan_active_forecast_export_task/grid

        - Plan & Archived forecast (adhoc) - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.test.exp-aws.net/dags/epm_vtrplan_cube_data_export_task/grid
    - **Airflow variables for both Prod and Test environments:**
        - **Active forecast:**
            - "vtr-plan-data-load-image-version" → docker image version (github commit sha)
            - "vtr-plan-active-forecast-scenario" → Forecast
            - "vtr-plan-active-forecast-version" → Working_Prior
            - "vtr-plan-active-forecast-years" → FY26,FY27
            - "vtr-plan-active-forecast-period" → Yeartotal
        - **Plan & Archived forecast (adhoc):**
            - "vtr-plan-data-load-image-version" → docker image version (github commit sha)
            - "vtr-plan-data-load-scenario" → Feb Forecast
            - "vtr-plan-data-load-version" → FY26_Archive
            - "vtr-plan-data-load-years" → FY26,FY27
            - "vtr-plan-data-load-period" → Yeartotal
- **Metadata dimensions**:
    - **Prod** - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.dw.exp-aws.net/dags/epm_metadata_export_task/grid
    - **Test** - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.test.exp-aws.net/dags/epm_metadata_export_task/grid
    - **Airflow variables for both Prod and Test environments:**
        - "epm-metadata-load-image-version" → docker image version (github commit sha)
- **Fxrates**:
    - Archived forecast rates will be updated as part of loading the Archived forecast data for any archived period. This will be run adhoc whenever the forecast is locked.
    - Working forecast rates will be updated daily twice as part of loading the daily active forecast data. The active forecast job will run twice a day at:
        - 1st run - 7:30 AM PST
        - 2nd run - 11:30 PM PST
    - But if we want to load/reload just the fxrates, then here are the airflow jobs:
        - **Prod** - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.dw.exp-aws.net/dags/epm_fxrates_export_task/grid
        - **Test** - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.test.exp-aws.net/dags/epm_fxrates_export_task/grid

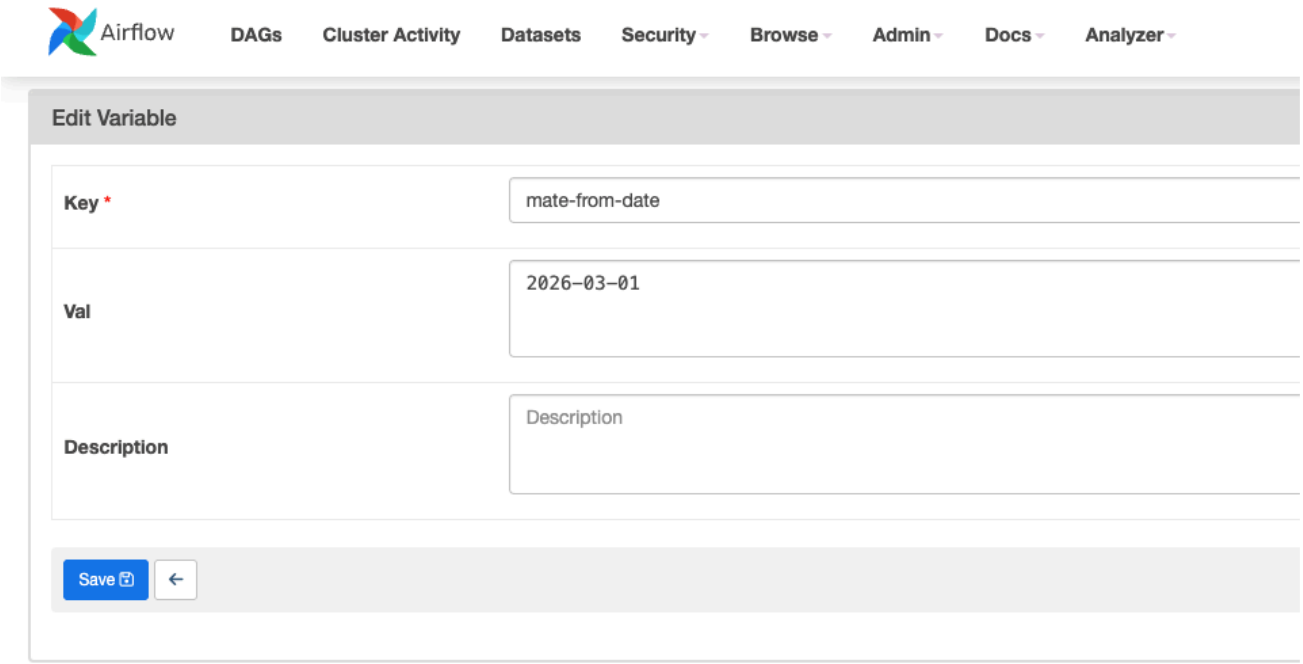
- **Airflow variables for both Prod and Test environments:**
  - `fxrates-load-image-version` → docker image version (github commit sha)
  - `fxrates-load-scenario` → Ex: Mar Forecast
  - `fxrates-load-version` → Ex: FY26_Archive

Steps to run an airflow job:

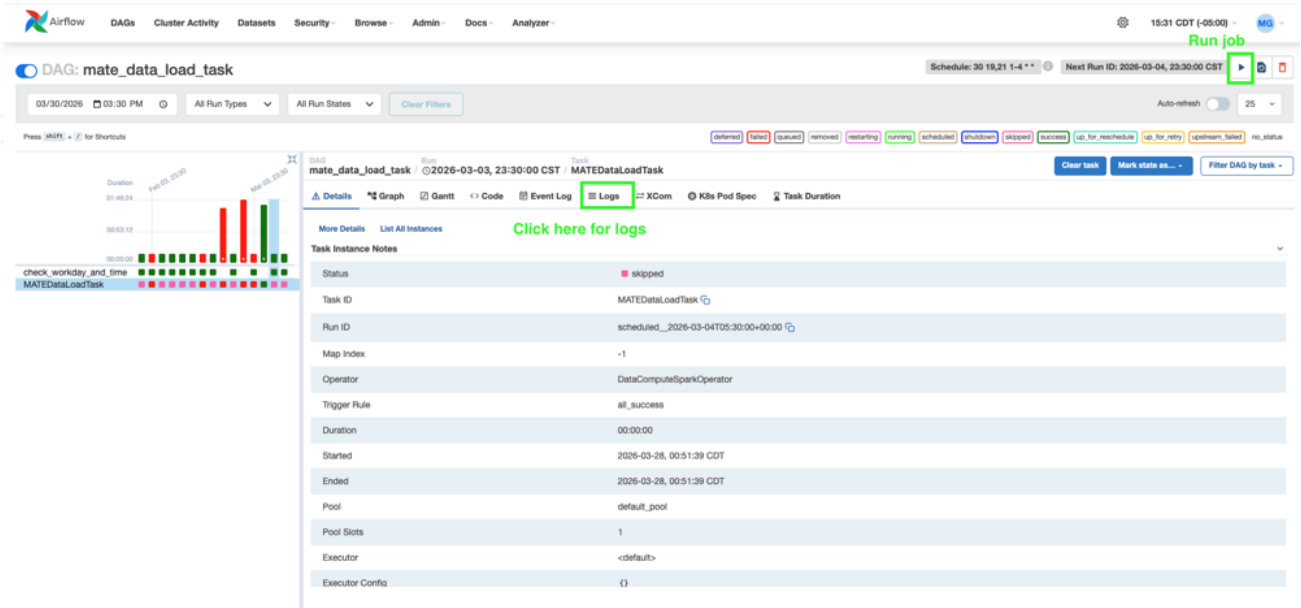
1. Navigate to the airflow job that we want to run based on above links or go to home page - https://airflow-finance-planning-forecasting.wo-eks-us-east-1.data.dw.exp-aws.net/home and click on the desired job.
2. From the top menu, click on Admin drop down and select Variables



3. Search for the variables that are required to run the desired job and update them as needed based on the criteria. We can update variables by clicking on the 'Edit Record' button to the left of the variable and saving it.



4. Now go back to the airflow job and click on Play button on the top right. Once the job starts, we can click on the Logs tab to see the airflow logs. We can also find Kubernetes dashboard link, splunk logs link and spark-ui link in the airflow logs for the job we ran.



Data sets and schedules:

All the below jobs are scheduled based on PST and shouldn't be affected by Day light savings.

FIN_PLAN – Cube where all sub-models (VTR and Workforce) will integrate their data and users will finalize their Forecasts.

VTR_PLAN – Cube where daily transaction level data will be loaded to calculate timing cards.

Data set	Cube	Criteria to fetch	Published	Schedule
Actuals	FIN_PLAN	scenario = Actual version = Final	After each consolidation monthly	Scheduled for WD1, WD2 & WD3 of each month at 11 PM PST. Adhoc - after the consolidation

Plan	FIN_PLAN & VTR_PLAN	scenario = Plan version = Final	After the Plan has been locked	Adhoc - Once after the Plan has been finalized
Finalized Forecast / Archived Forecast	FIN_PLAN & VTR_PLAN	scenario = <MMM> Forecast (Ex: Jan Forecast) version = <FYxx>_Archive (Ex: FY25_Archive)	After the current month forecast has been locked	Adhoc - Once a month after the forecast has been finalized
Active Forecast / Working Forecast	FIN_PLAN & VTR_PLAN	scenario = Forecast version = Working_Prior	Ongoing basis - daily	Scheduled daily at: • 1st run - 7:30 AM PST • 2nd run - 11:30 PM PST
Dimensions Metadata		from dimension tables above	Daily	Scheduled daily at 11 PM PST
Fxrates	FIN_PLAN	• Archived forecast rates: ◦ scenario = <MMM> Forecast (Ex: Jan Forecast) version = <FYxx>_Archive (Ex: FY26_Archive) • Working forecast rates: ◦ scenario = NoScenario	Archived forecast rates - adhoc → Whenever archived forecast is loaded on a monthly basis Working forecast rates - Daily Spot rates → Whenever actuals are loaded on a monthly basis	• Archived forecast rates will be updated as part of loading the Archived forecast data for any locked period. This will be run adhoc whenever

		◦ version = NoVersion • Spot rates: ◦ scenario = Actual ◦ version = Final		the forecast is locked. • Spot rates will be loaded as part of loading the Actuals data for a period. This will be run whenever the actuals are loaded during the first few working days of the month. • Working forecast rates will be updated daily twice as part of loading the daily active forecast data. The active forecast job will run twice a day like mentioned above at: • 1st run - 7:30 AM PST

				• 2nd run - 11:30 PM PST

EGDL tables:

Use **egdp_test_finance_analytics** schema for test environment and **egdp_prod_finance_analytics** for prod environment.

- **Metadata tables -**
  - ```
    select * from egdp_prod_finance_analytics.epm_account_metadata_table;
    select * from egdp_prod_finance_analytics.epm_brand_metadata_table;
    select * from egdp_prod_finance_analytics.epm_businessmodel_metadata_table;
    select * from egdp_prod_finance_analytics.epm_channel_metadata_table;
    select * from egdp_prod_finance_analytics.epm_company_metadata_table;
    select * from egdp_prod_finance_analytics.epm_costcenter_metadata_table;
    select * from egdp_prod_finance_analytics.epm_currency_metadata_table;
    select * from egdp_prod_finance_analytics.epm_day_metadata_table;
    select * from egdp_prod_finance_analytics.epm_hsp_metric_metadata_table;
    select * from egdp_prod_finance_analytics.epm_intercompany_metadata_table;
    select * from egdp_prod_finance_analytics.epm_lineitem_metadata_table;
    select * from egdp_prod_finance_analytics.epm_location_metadata_table;
    select * from egdp_prod_finance_analytics.epm_period_metadata_table;
    select * from egdp_prod_finance_analytics.epm_planelement_metadata_table;
    select * from egdp_prod_finance_analytics.epm_product_metadata_table;
    select * from egdp_prod_finance_analytics.epm_scenario_metadata_table;
    select * from egdp_prod_finance_analytics.epm_sec_attribute_metadata_table;
    select * from egdp_prod_finance_analytics.epm_smart_lists_metadata_table;
    select * from egdp_prod_finance_analytics.epm_timingcard_metadata_table;
    select * from egdp_prod_finance_analytics.epm_version_metadata_table;
    select * from egdp_prod_finance_analytics.epm_view_metadata_table;
    select * from egdp_prod_finance_analytics.epm_years_metadata_table;
    ```
- **Cube data tables -**
  - ```
    select * from egdp_prod_finance_analytics.epm_fin_plan_cube_table;
    select * from egdp_prod_finance_analytics.epm_vtr_plan_cube_table;
    ```
- **Fxrates table**
  - ```
    select * from egdp_prod_finance_analytics.epm_fxrates_table
    ```

Implementation details:

Github repo - https://github.expedia.biz/accounting-data-engineering/epm-data-processing

- fin_plan cube load → https://github.expedia.biz/accounting-data-engineering/epm-data-processing/blob/main/module/src/main/scala/com/expediagroup/data/task/FINPlanCubeDataLoadTask.scala
    ```

- vtr_plan cube load → https://github.expedia.biz/accounting-data-engineering/epm-data-processing/blob/main/module/src/main/scala/com/expediagroup/data/task/VTRPlanCubeDataLoadTask.scala
- dimensions metadata load → https://github.expedia.biz/accounting-data-engineering/epm-data-processing/blob/main/module/src/main/scala/com/expediagroup/data/task/EPMMetaDataLoadTask.scala
- fxrates load → https://github.expedia.biz/accounting-data-engineering/epm-data-processing/blob/main/module/src/main/scala/com/expediagroup/data/task/FxratesDataLoadTask.scala

FIN_PLAN Dimensionality

Click here to expand...

Dimension	Description	Additional Details
Years	The Years dimension represents the various plans and actual years where data will be stored.	
Period		
Account		
Brand		
BusinessModel		
Channel		
Company		

CostCenter		
Currency		
HSP_View	The HSP_View dimension created when a cube is enabled for the Sandboxes feature, the following members is created: BaseData is the default member where data is stored when users are NOT working in a sandbox. SandboxData is the place where a user works with data in a sandbox. ConsolidatedData is a dynamically calculated member and will retrieves data from the SandboxData member if it's available. Otherwise, the ConsolidatedData member retrieves data from the BaseData member of the base version.	
LineItem		
Location		
PlanElement		
Product		
Scenario		
Version		

Key Benefits of the New Approach:

1. **Automated & Scalable Data Flow**
   - Data movement will be fully **automated** using **scheduled jobs, Oracle Cloud Integrations, and AWS S3**, reducing manual efforts.
2. **Enhanced Security & Controlled Access**
   - Ensures **secure** data transfer with **encryption** in transit and at rest.
   - Role-based **access controls** will regulate data consumption.
   - Audit logs will provide **visibility** into data access and modifications.
3. **Advanced Analytics & Cross-Platform Accessibility**
   - **Standardized data availability** in EGDL enables teams to perform advanced analytics without manual data wrangling.
   - Supports integration with BI tools, ML models, and other analytical platforms.
4. **Compliance & Governance**
   - Adheres to Expedia Group's **data security policies** and **governance standards** to ensure regulatory compliance.

Conclusion:

- This transformation **streamlines data accessibility**, enhances security, and ensures consistency across teams.
- By leveraging EGDL, we enable **faster decision-making** and **better collaboration** across the organization.

Requirements Discussion Recording

Programmatic Access.mp4

Questions and Discussions

Item	Details	Created By	Assigned To/Team	Response	Status

1.	Do we need to publish Exchange Rates	@Piyoosh Pant	@Nick Momenah	Yes we would need to publish the rates including the constant currency rates.	CLOSED
2.	De we need to publish the Dimension Hierarchy	@Piyoosh Pant	Integration Team	We would need to publish the hierarchy possibly from	OPEN
3.	Restatement - What will be the strategy to restate plan/forecast in the system. The Plan and Forecast will have to be republished to EGDL when we have a restatement.	@Piyoosh Pant	@Ryan Ward @Nate Bannick	We will have to publish restated Plan and Forecast numbers in EGDL to sync with the system - Nate to review	OPEN
4.	Open forecast - Need some way to display if the forecast is active or locked. A variable that could display the status and be published as a column in the EGDL	@Piyoosh Pant	Integration Team		OPEN
5.	For any data published to EGDL it would be helpful if we have the published timestamp added to each dataset	@Piyoosh Pant	Integration Team		OPEN
6.	Security and data partitioning requirement	@Piyoosh Pant	@Abhishek Marathe @Nate Bannick	Abhishek & Nate to provide access skeleton by EOW	OPEN

7.	System Performance - Do we have any system performance considerations when we are extracting data from the system	@Piyoosh Pant	@Ryan Ward		OPEN
8.	The GFT Product/Program team will collaborate with all downstream consumers of Hyperion data to educate them on the new Chart of Accounts. Until the hard launch, they will continue receiving data from the on-prem system. All downstream systems must be retrofitted to consume data from EGDL using the new Chart of Accounts before the hard cutover.	@Piyoosh Pant	@Abhishek Marathe @Nate Bannick	No action needed, Education to based on guidance from Nick and Nate - Nate to review	OPEN

9	• We will be loading data from 2 cubes - FIN_PLAN and VTR_PLAN.
	• Monthly actual data will only be for FIN_PLAN. Finalized Forecast, Plan and Active Forecast will be for both FIN_PLAN and VTR_PLAN.
	• 4 airflow jobs in total -
	  ◦ Active Forecast for FIN_PLAN - scheduled daily
	  ◦ Active Forecast for VTR_PLAN - scheduled daily
	  ◦ Finalized Forecast, Plan and Monthly Actual for FIN_PLAN - run manually as needed by updating airflow variables accordingly
		@Manoj Gangula	@Ryan Ward @Piyoosh Pant	Agreed on these points.	

- Finalized Forecast, Plan and Monthly Actual for FIN_PLAN - run manually as needed by updating airflow variables accordingly
- Airflow jobs will accept 4 variables for scenario, version, years and period.
- We will overwrite the data based on partition criteria which will be (scenario,version,years,period)
  - Monthly actual job will only overwrite the same month data if we are running it multiple times during a month. It will not overwrite previous month data since period will be different.
  - Likewise for the other jobs, data will be overwritten only if it matches the above partition criteria
- BegBalance will be fetched for all the above jobs.
- Groovy script for copying data from Working to Working_Prior version will be run only for Active Forecast case.
- Data load criteria from cubes -
  - Account
    - FIN_PLAN - Lvl0descendants of NONFIN, INCMST, Driver_Rates_stats
    - VTR_PLAN - Lvl0descendants of

Booked_To_Stayed_Metrics

- Currency - Local_Currency, USD_Constant, USD_GL, USD_Constant_Orig
- LineItem
  - FIN_PLAN - Total_LineItem (No Lvl0descendants)
  - VTR_PLAN - NoLineItem
- HSP_View - BaseData
- Company - Lvl0descendants of All_LedgerCurrency and Timing_Card_LE
- PlanElement -
  - FIN_PLAN - Total_Plan, Total_Plan_w_CorpFPA_Reserve (No Lvl0descendants)
  - VTR_PLAN - Stayed_Value_or_Amt, Timing_Card_Default_Pct, Timing_Card_Forecast_Applied_Pct (No Lvl0descendants)
- BusinessModel - ILvl0Descendants(Total_BusinessModel)
- Brand - ILvl0Descendants(Total_Brand)
- Channel - ILvl0Descendants(Total_Channel)
- Location - ILvl0Descendants(Total_Location)
- Product - ILvl0Descendants(Total_Prod

	uct) ◦ CostCenter ▪ FIN_PLAN - ILvl0Descendants(Total_CostCenter) ◦ Years - from airflow variable ◦ Scenario - from airflow variable ◦ Version - from airflow variable ◦ Period - BegBalance, <period> passed from airflow variables ◦ View - ▪ VTR_PLAN - Periodic ◦ Day - ▪ VTR_PLAN - No_Day ◦ TimingCard - ▪ VTR_PLAN - ILvl0Descendants(Current_TimingCard)				
	• Updated PlanElement for FIN_PLAN - ◦ Actuals - ILvl0Descendants(Actuals_Detail),Actuals_Allocations ◦ Forecast & Plan - ILvl0Descendants(Total_Plan),CorpFPA_Adjustment • We will fetch ILvl0Descendants of Actuals_Detail and Total_Plan, so the plan element column in EGDL will show all the respective plan elements. We will let the consumers look at the data and then we can	@Manoj Gangula	@Ryan Ward @Abhishek Marathe @Piyoosh Pant @Protim		

	consolidate everything to a single plan element like Total_Plan based on the feedback. • When we fetch ILvl0Descendants(Actuals_Detail), all the plan elements under this parent including the negates will be retrieved. Users should be able to filter these by selecting the plan element they need. We can update these based on the users feedback and use cases.				
	OKC - Loyality accounts ([FSIT-166797] Create new stat accounts for Loyalty Data - Expedia Jira) These new accounts are setup in Hyperion to support OKC Dashboard and need to be part of the data that is being published to EGDL. @Manoj Gangula @Abhishek Marathe • Loyalty Pgrm Earn - PDRP Loyalty Cost (SR0708) • Loyalty Pgrm Earn - PDRP Claims Cost (SR0709) • Loyalty Pgrm Earn - OK Earn Initiatives (SR0710) • Loyalty Pgrm Cost - Targeted CRM (SR0711)				